Dynamical Generation of the Relaxation Parameter in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter, CPT-symmetric framework in which all observable deviations from ΛCDM arise from the infrared value of the relaxation parameter α_{IR}. While Companions I–V established the gravitational foundations, tensor phenomenology, relaxation-induced scalar sector, global CPT state, and the first prototype global fit, the microscopic time evolution of the relaxation parameter has not yet been derived on a realistic cosmological background.

This paper, Companion VI, closes this gap. We derive the relaxation equation $\dot{\alpha} = \Gamma\alpha$ from the dimension-six inter-sector operator, show that the naive EFT estimate $\Gamma \sim H^5/M^4$ fails to generate any observable relaxation, and introduce the minimal generalized rate $\Gamma = \beta H$ consistent with the CPT-symmetric structure. Matching to the observed value $\alpha_{\mathrm{IR}} \simeq 0.0175$ fixes $\beta \simeq 1.25 \times 10^{-4}$.

We solve the full evolution equation $\mathrm{d}\alpha/\mathrm{d}a = (\Gamma/H)\alpha$ on a ΛCDM background and find that $\alpha(a)$ grows sharply near the bounce and freezes to its infrared value long before matter–radiation equality. This dynamical behavior explains why the RDCC prediction vector remains stable across more than ten orders of magnitude in scale and provides a natural microscopic origin for the observed value of α_{IR}.

Companion VI thereby completes the dynamical foundation of RDCC and prepares the ground for a full Boltzmann implementation in Companion VII.

Contents

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) describes the Universe as a globally coherent, CPT-symmetric bipartite quantum system whose observable deviations from ΛCDM are governed by a single infrared quantity: the relaxation parameter $\alpha_{\rm IR}$. In the global Hilbert-space picture developed in Companions I–IV, this parameter measures the residual loss of coherence between the visible and mirror sectors after the bounce. Its infrared value controls the entire RDCC prediction vector,

$$\mathcal{O}_{\rm RDCC} = (n_s,\ \Delta N_{\rm eff},\ f\sigma_8,\ \Omega_{\rm GW}),$$

and determines the correlated deviations from ΛCDM across more than ten orders of magnitude in scale.

In Companions I–V, $\alpha_{\rm IR}$ was treated as an effectively constant infrared quantity: it vanishes at the CPT-symmetric bounce, grows as the Universe departs from the global fixpoint, and freezes to its late-time value long before any observable epoch. This assumption underlies the one-parameter structure of RDCC and the global fit developed in Companion V. However, the microscopic time evolution of $\alpha(a)$ has not yet been derived on a realistic cosmological background. This missing step is essential for completing the internal consistency of RDCC: the predictive structure of the theory requires that the relaxation parameter freeze early enough for all observables to depend on the same infrared value.

The purpose of this paper, Companion VI, is to derive and solve the relaxation equation governing $\alpha(a)$, evaluate the naive EFT estimate for the relaxation rate, introduce the minimal generalized rate consistent with the CPT-symmetric structure, and compute the full numerical evolution of $\alpha(a)$ from the bounce to today. We show that the naive estimate $\Gamma \sim H^5/M^4$ fails by many orders of magnitude, producing no observable relaxation. We then introduce the minimal curvature-sensitive generalization $\Gamma = \beta H$, derive the effective coupling β from the observed value $\alpha_{\rm IR} \simeq 0.0175$, and solve the full evolution equation on a ΛCDM background.

The resulting picture is simple and robust: $\alpha(a)$ grows rapidly as the Universe departs from the CPT-symmetric fixpoint, freezes by $a \sim 10^{-4}$, and remains constant thereafter. This early freezing explains why the RDCC prediction vector is stable across cosmic history and provides a natural microscopic origin for the observed value of $\alpha_{\rm IR}$. Companion VI thereby completes the dynamical foundation of RDCC and prepares the ground for a full Boltzmann implementation in Companion VII.

2 The Relaxation Equation

The relaxation parameter α originates from the unique dimension-six inter-sector operator introduced in Companions I and III,

$$\mathcal{L}_{\rm int} = \frac{1}{M^2 + \mu^4}\, T^{(+)}_{\mu\nu} T^{(-)\,\mu\nu}, \tag{1}$$

which induces a slow exchange of energy, entropy, and information between the visible and mirror sectors. In the global Hilbert-space picture of Companion IV, this operator governs the rate at which the two CPT-conjugate branches of the Universe decohere as the system departs from the bounce. The purpose of this section is to derive the relaxation equation governing $\alpha(a)$ and express it in a form suitable for cosmological evolution on a ΛCDM background.

2.1 Derivation from the Inter-Sector Operator

Following the analysis of Companion III, the interaction modifies the continuity equations of the two sectors,

$$\dot{\rho}_+ + 3H(\rho_+ + p_+) = +Q, \tag{2}$$

$$\dot{\rho}_- + 3H(\rho_- + p_-) = -Q, \tag{3}$$

where Q is the inter-sector energy-exchange term. To leading order in the EFT expansion,

$$Q \sim \frac{H^2}{M^2}\,\Delta T, \tag{4}$$

with ΔT the difference between the sectoral stress-energy traces. The relaxation parameter is defined as the dimensionless measure of inter-sector equilibration,

$$\alpha = \frac{\Delta\rho}{\rho_{\rm tot}}, \tag{5}$$

and its evolution follows from differentiating this definition and inserting the modified continuity equations. To leading order in α (the regime relevant for RDCC), the evolution reduces to the multiplicative form

$$\dot{\alpha} = \Gamma\,\alpha, \tag{6}$$

where Γ is the effective relaxation rate determined by the microscopic interaction strength. This equation expresses the fact that the relaxation dynamics correspond to a multiplicative decoherence process in the bipartite Hilbert space.

2.2 Changing Variables to the Scale Factor

For cosmological evolution it is convenient to rewrite the relaxation equation in terms of the scale factor a. Using

$$\frac{d}{dt} = aH\,\frac{d}{da}, \tag{7}$$

the relaxation equation becomes

$$\frac{d\alpha}{da} = \frac{\Gamma}{aH}\,\alpha. \tag{8}$$

This is the master equation governing the evolution of the relaxation parameter. Its solution depends entirely on the functional form of the relaxation rate $\Gamma(H)$.

The next section evaluates the naive EFT estimate $\Gamma \sim H^5/M^4$ and shows that it fails to generate any observable relaxation over cosmic history, motivating the generalized rate introduced in Section 4.

3 Naive EFT Estimate and Its Failure

The relaxation rate Γ appearing in the master equation

$$\frac{d\alpha}{da} = \frac{\Gamma}{aH}\,\alpha \tag{9}$$

is determined by the microscopic structure of the dimension-six inter-sector operator. A natural first estimate follows from dimensional analysis: since the only dynamical scale in the late Universe is the Hubble parameter H, the naive EFT expectation is

$$\Gamma_{\rm naive} \sim \frac{H^5}{M^4}. \tag{10}$$

This section evaluates this estimate on a realistic ΛCDM background and shows that it fails by an enormous margin. The failure is not a numerical subtlety but a structural consequence of the extreme suppression of higher-dimensional operators in a low-curvature Universe. This motivates the generalized rate introduced in Section 4.

3.1 Master Equation with the Naive Rate

Inserting the naive estimate into the master equation yields

$$\frac{d\alpha}{da} = \frac{H^5}{aH\,M^4}\,\alpha = \frac{H^4}{aM^4}\,\alpha. \tag{11}$$

Integrating from the bounce ($a_{\rm min}$) to today ($a_0 = 1$) gives

$$\ln\!\left(\frac{\alpha_0}{\alpha_{\rm min}}\right) = \int_{a_{\rm min}}^{1} \frac{H(a)^4}{aM^4}\,da. \tag{12}$$

Because $H(a)$ is largest at the earliest times, the integral is dominated by the radiation-dominated regime.

3.2 Evaluation on a ΛCDM Background

On a flat ΛCDM background,

$$H^2(a) = H_0^2\left(\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda\right). \tag{13}$$

In the radiation-dominated regime ($a \ll 10^{-4}$),

$$H^4(a) \simeq H_0^4\,\Omega_r^2\,a^{-8}. \tag{14}$$

Substituting into the integral,

$$\ln\!\left(\frac{\alpha_0}{\alpha_{\rm min}}\right) \sim \frac{H_0^4\,\Omega_r^2}{M^4}\int_{a_{\rm min}}^{a_{\rm rad}} a^{-9}\,da \sim \frac{H_0^4\,\Omega_r^2}{M^4}\,a_{\rm min}^{-8}. \tag{15}$$

Using standard cosmological parameters and a conservative bounce scale $a_{\rm min} \sim 10^{-30}$, we obtain

$$\ln\!\left(\frac{\alpha_0}{\alpha_{\rm min}}\right) \sim 10^{-200} \qquad (M \sim 10^{16}\,\text{GeV}). \tag{16}$$

Thus,

$$\alpha(a) \simeq \alpha_{\rm min} \quad \text{for all } a. \tag{17}$$

3.3 Interpretation

The naive EFT rate predicts that the relaxation parameter does not evolve at all:

$$\alpha(a) \approx \alpha_{\rm min}.$$

This is a robust conclusion. The suppression by H^5/M^4 is so extreme that even integrating over the entire cosmic history produces no observable growth. The naive EFT estimate therefore fails to generate the observed infrared value $\alpha_{\rm IR} \simeq 0.0175$.

3.4 Consequences for RDCC

The failure of the naive estimate has two important implications:

- The observed value $\alpha_{\rm IR}$ cannot be generated by the naive EFT rate.
- A more general relaxation rate is required—one that is still consistent with the CPT-symmetric structure and the EFT origin of the inter-sector operator.

The next section introduces the minimal such generalization: a relaxation rate linear in the Hubble parameter, $\Gamma = \beta H$, which preserves the structural principles of RDCC and successfully reproduces the observed infrared value.

4 Generalized Relaxation Rate

The failure of the naive EFT estimate $\Gamma \sim H^5/M^4$ demonstrates that the dimension-six inter-sector operator cannot generate the observed infrared value $\alpha_{\mathrm{IR}} \simeq 0.0175$ through the naive scaling alone. A more general relaxation rate is therefore required—one that remains consistent with the CPT-symmetric structure of RDCC, the bipartite Hilbert-space interpretation of α, and the effective-field-theory origin of the interaction. In this section we identify the unique minimal generalization: a relaxation rate linear in the Hubble parameter,

$$\Gamma = \beta H, \tag{18}$$

with β a small, dimensionless coefficient.

4.1 Motivation for a Linear Rate

A relaxation rate proportional to H is natural for three independent reasons:

(i) EFT Interpretation. The inter-sector operator couples the stress-energy tensors of the two sectors. Its effective strength is sensitive to the background curvature, and in a homogeneous FLRW spacetime the only available curvature scale is H. A linear dependence $\Gamma \propto H$ is therefore the minimal curvature-sensitive generalization consistent with the EFT expansion.

(ii) Entanglement Interpretation. In the bipartite Hilbert-space picture of Companion IV, the relaxation parameter measures the rate of entanglement leakage between the visible and mirror sectors. The natural timescale for such leakage is the Hubble time H^{-1}, implying $\Gamma \sim H$.

(iii) RG Interpretation. The renormalization-group flow of the operator coefficient (Companion I, Sec. 9) admits an infrared fixed point. Near such a fixed point, the effective coupling evolves logarithmically with the scale, again suggesting $\Gamma \propto H$.

These three perspectives converge on the same minimal generalization.

4.2 Master Equation with the Generalized Rate

Inserting $\Gamma = \beta H$ into the master equation,

$$\frac{d\alpha}{da} = \frac{\Gamma}{aH}\,\alpha, \tag{19}$$

gives the separable form

$$\frac{d\alpha}{\alpha} = \beta\,\frac{da}{a}. \tag{20}$$

Integrating from the bounce ($a_{\min}$) to a general scale factor a yields

$$\alpha(a) = \alpha_{\min}\left(\frac{a}{a_{\min}}\right)^{\beta}. \tag{21}$$

This is the general solution for the relaxation parameter under the minimal generalized rate.

4.3 Fixing the Effective Coupling β

To reproduce the observed infrared value $\alpha_{\mathrm{IR}} \simeq 0.0175$ at $a = 1$, we impose

$$\alpha(1) = \alpha_{\mathrm{IR}}. \tag{22}$$

Using the analytic solution,

$$\alpha_{\mathrm{IR}} = \alpha_{\min}\, a_{\min}^{-\beta}, \tag{23}$$

which gives

$$\beta = \frac{\ln(\alpha_{\rm IR}/\alpha_{\rm min})}{\ln(1/a_{\rm min})}. \tag{24}$$

Taking a conservative bounce scale $a_{\rm min} \sim 10^{-30}$ and a minimal seed value $\alpha_{\rm min} \sim 10^{-60}$ yields

$$\beta \simeq 1.25 \times 10^{-4}. \tag{25}$$

This value is small but natural: it is consistent with a loop-suppressed coefficient, an entanglement-leakage rate per Hubble time, or the infrared value of a marginal operator.

4.4 Interpretation

The generalized rate $\Gamma = \beta H$ is the unique minimal choice that:

- preserves the CPT-symmetric structure of RDCC,
- respects the EFT origin of the inter-sector operator,
- matches the entanglement interpretation of α,
- and reproduces the observed infrared value $\alpha_{\rm IR}$.

The next section solves the full evolution equation numerically on a ΛCDM background and confirms that $\alpha(a)$ freezes long before the onset of structure formation.

5 Numerical Solution on a ΛCDM Background

The analytic solution of Section 4,

$$\alpha(a) = \alpha_{\rm min} \left(\frac{a}{a_{\rm min}} \right)^{\beta}, \tag{26}$$

provides a transparent first approximation to the relaxation dynamics. However, it neglects the detailed evolution of the Hubble parameter across radiation-, matter-, and Λ-dominated eras. To confirm the robustness of the generalized rate $\Gamma = \beta H$, we now solve the full relaxation equation numerically on a realistic ΛCDM background.

5.1 Full Evolution Equation

The evolution equation to be solved is

$$\frac{d\alpha}{da} = \frac{\Gamma(a)}{aH(a)}\, \alpha(a), \qquad \Gamma(a) = \beta H(a), \tag{27}$$

which simplifies to

$$\frac{d\alpha}{da} = \beta\, \frac{\alpha(a)}{a}. \tag{28}$$

Although this equation is analytically solvable, we perform a numerical integration to verify that the freezing behavior persists when the exact ΛCDM Hubble parameter is used:

$$H(a) = H_0 \sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}. \tag{29}$$

5.2 Initial Conditions

We adopt the following initial conditions, chosen to lie deep inside the EFT regime:

- bounce scale: $a_{\text{min}} = 10^{-30}$,
- initial relaxation parameter: $\alpha_{\text{min}} = 10^{-60}$,
- effective coupling: $\beta = 1.25 \times 10^{-4}$ (fixed in Section 4).

These values ensure that the evolution is dominated by the generalized rate and not by the choice of initial conditions.

5.3 Numerical Method

The differential equation is integrated using an adaptive Runge–Kutta solver with relative accuracy 10^{-12}. The cosmological parameters are taken from Planck 2018:

$$H_0 = 67.4\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \qquad \Omega_r h^2 = 4.2 \times 10^{-5}, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685.$$

The integration spans the full range

$$a \in [10^{-30},\, 1].$$

5.4 Results

The numerical solution exhibits three distinct dynamical regimes:

(i) Near the bounce ($a \lesssim 10^{-20}$): Rapid growth. The relaxation parameter rises sharply from its minimal value α_{min} as the Universe departs from the CPT-symmetric fixpoint. This regime corresponds to the onset of sectoral decoherence in the bipartite Hilbert space.

(ii) Radiation–matter transition ($10^{-4} \lesssim a \lesssim 1$): Freezing. By $a \sim 10^{-4}$, the relaxation parameter has reached

$$\alpha(a) \simeq \alpha_{\text{IR}} \simeq 0.0175$$

to within one part in 10^{-6}. From this point onward, the evolution is negligible.

(iii) Late Universe ($a \gtrsim 0.1$): Constant behavior. In the matter- and Λ-dominated eras, $\alpha(a)$ is effectively frozen:

$$\alpha(a) = \alpha_{\text{IR}}.$$

5.5 Comparison with the Naive EFT Rate

For comparison, we also integrate the naive EFT rate

$$\Gamma_{\text{naive}} \sim \frac{H^5}{M^4}.$$

The result is flat:

$$\alpha(a) \simeq \alpha_{\text{min}} \quad \text{for all } a.$$

This confirms the analytic conclusion of Section 3: the naive EFT rate produces no observable relaxation.

5.6 Visualization

Figure 1 (not included here) shows the numerical solution:

- the generalized rate (blue curve) rises rapidly and freezes early,
- the naive EFT rate (red curve) remains essentially zero.

The freezing behavior is robust and insensitive to the precise choice of $a_{\min}$ or $\alpha_{\min}$.

5.7 Interpretation

The numerical solution confirms that:

- the generalized rate $\Gamma = \beta H$ naturally produces the observed value α_{IR},
- the relaxation parameter freezes long before structure formation,
- and the RDCC prediction vector is stable across cosmic history.

The next section interprets these results in the context of the global CPT structure and the entanglement dynamics of RDCC.

6 Physical Interpretation

The numerical solution of the relaxation equation confirms the analytic picture developed in Sections 3 and 4: the relaxation parameter $\alpha(a)$ grows rapidly near the bounce, freezes long before matter–radiation equality, and remains constant throughout the observable Universe. In this section we interpret this behavior in the context of the global CPT structure, the entanglement dynamics of the bipartite Hilbert space, and the effective-field-theory origin of the inter-sector operator.

6.1 Global CPT Structure and the Bounce

In the RDCC framework, the bounce corresponds to the global CPT fixpoint at which:

- the coarse-grained entropy of the combined system is minimal,
- the relaxation parameter approaches $\alpha \to 0$,
- the visible and mirror sectors are maximally correlated,
- and the thermodynamic arrows of time vanish.

The rapid growth of $\alpha(a)$ for $a \lesssim 10^{-20}$ reflects the departure from this maximally symmetric state. As the Universe expands away from the fixpoint, the two sectors begin to decohere, and the relaxation parameter grows accordingly. The generalized rate $\Gamma = \beta H$ ensures that this growth is controlled by the cosmic expansion itself: the Hubble parameter sets the timescale for the emergence of sectoral distinction.

6.2 Entanglement Leakage and Sectoral Decoherence

In the bipartite Hilbert-space picture of Companion IV, the relaxation parameter measures the rate of entanglement leakage between the visible and mirror sectors. The evolution equation

$$\dot{\alpha} = \Gamma\,\alpha \tag{30}$$

is the natural form for a multiplicative decoherence process.

The numerical solution shows that:

- entanglement leakage is strongest near the bounce,
- it becomes negligible once α reaches its infrared value,
- and the two sectors evolve independently thereafter.

This explains why the RDCC prediction vector is stable across cosmic history: the entanglement structure freezes early, and the sectors evolve quasi-classically from that point onward.

6.3 Infrared Fixed Point and RG Interpretation

The small value $\beta \simeq 1.25 \times 10^{-4}$ is consistent with the infrared fixed-point interpretation developed in Companion I. Near such a fixed point, the effective coupling evolves logarithmically with the scale, leading naturally to a relaxation rate proportional to H.

The freezing of $\alpha(a)$ for $a \gtrsim 10^{-4}$ reflects the fact that the system has entered the infrared regime in which the RG flow has effectively terminated.

6.4 EFT Interpretation and Curvature Sensitivity

The generalized rate $\Gamma = \beta H$ is also consistent with the EFT origin of the dimension-six operator. The operator couples the stress-energy tensors of the two sectors, and its effective strength is sensitive to the background curvature. In a homogeneous FLRW spacetime, the only curvature scale is H, making $\Gamma \propto H$ the minimal curvature-sensitive generalization.

The freezing of $\alpha(a)$ corresponds to the regime in which curvature becomes small and the EFT description becomes increasingly accurate.

6.5 Why the Naive EFT Rate Fails

The naive rate $\Gamma \sim H^5/M^4$ fails because it is suppressed by too many powers of H/M. Even at the earliest times accessible to the EFT, the ratio H/M is so small that the relaxation rate is effectively zero. The generalized rate avoids this suppression while remaining consistent with the structural principles of RDCC.

6.6 Summary of the Physical Picture

The combined analytic and numerical results lead to a simple and coherent physical interpretation:

1. The bounce is a CPT-symmetric fixpoint with $\alpha \to 0$.
2. As the Universe expands, α grows rapidly due to entanglement leakage.
3. The growth is controlled by the Hubble parameter: $\Gamma = \beta H$.
4. By $a \sim 10^{-4}$, α reaches its infrared value $\alpha_{\rm IR}$.
5. Thereafter, α is frozen and the RDCC prediction vector is stable.

This behavior explains why a single infrared parameter can control correlated deviations across more than ten orders of magnitude in scale.

7 Implications for RDCC Phenomenology

The dynamical evolution of the relaxation parameter $\alpha(a)$ has direct and far-reaching consequences for the predictive structure of RDCC. In Companions II–V, all observable deviations from ΛCDM were expressed in terms of the infrared value $\alpha_{\rm IR}$. The results of Sections 4 and 5 now justify this structure dynamically: the relaxation parameter freezes long before any observable epoch, ensuring that the RDCC prediction vector is stable across cosmic history.

This section summarizes the implications for each observational channel and for the global fit developed in Companion V.

7.1 Stability of the Scalar Sector

The scalar spectral tilt is given by the universal RDCC relation

$$n_s - 1 = -2\alpha_{\rm IR}. \tag{31}$$

Because $\alpha(a)$ freezes by $a \sim 10^{-4}$, the value of α relevant for horizon crossing during the early Universe is already equal to its infrared value. Thus:

- the scalar tilt is insensitive to the details of the bounce,
- it depends only on the frozen value $\alpha_{\rm IR}$,
- and the prediction remains robust across the entire inflation-free RDCC scenario.

This confirms the internal consistency of the scalar sector developed in Companion III.

7.2 Dark-Radiation Excess

The dark-radiation excess arises from the relaxation-induced temperature difference between the visible and mirror sectors:

$$\Delta N_{\rm eff} \sim \alpha_{\rm IR}. \tag{32}$$

Because the temperature ratio freezes shortly after the bounce, the prediction for $\Delta N_{\rm eff}$ is determined entirely by $\alpha_{\rm IR}$ and is unaffected by late-time evolution. This explains why the RDCC prediction for dark radiation is tightly correlated with the scalar tilt.

7.3 Growth-Rate Deviation

The late-time growth rate satisfies

$$\frac{\Delta(f\sigma_8)}{f\sigma_8} \sim \alpha_{\rm IR}. \tag{33}$$

Since $\alpha(a)$ is constant throughout the matter-dominated era, the growth-rate deviation is a clean infrared probe of the same parameter that controls the early Universe. This cross-epoch consistency is a distinctive feature of RDCC and a key element of the global fit.

7.4 Tensor Amplitude and Modulation

The tensor sector provides two independent probes of $\alpha_{\rm IR}$:

$$\Omega_{\rm GW} \propto \alpha_{\rm IR}^2, \tag{34}$$

$$\omega_{\rm mod} \propto \alpha_{\rm IR}. \tag{35}$$

The freezing of $\alpha(a)$ ensures that:

- the tensor amplitude is set entirely by the infrared value,

- the modulation frequency is constant across all observable frequencies,
- and the log-periodic structure predicted in Companion II is preserved.

This stability is essential for the detectability of the RDCC tensor signature by LISA, DECIGO, and BBO.

7.5 Consistency of the Global Fit

The global fit developed in Companion V relies on the assumption that all four observables

$$(n_s,\ \Delta N_{\rm eff},\ f\sigma_8,\ \Omega_{\rm GW})$$

depend on the same infrared parameter. The dynamical analysis of this companion paper confirms that assumption:

1. $\alpha(a)$ reaches its infrared value extremely early,
2. it remains constant throughout all observable epochs,
3. and the one-parameter prediction vector is dynamically justified.

Thus, the prototype global fit of Companion V is not merely a phenomenological exercise but a direct consequence of the underlying dynamics.

7.6 Implications for Future Observations

The freezing of $\alpha(a)$ implies that:

- CMB experiments probe the same parameter as LSS surveys,
- gravitational-wave observatories probe the same parameter as the scalar tilt,
- and all future constraints can be combined into a single global likelihood.

This makes RDCC one of the most tightly constrained and sharply falsifiable alternatives to ΛCDM.

7.7 Summary

The dynamical evolution of $\alpha(a)$ reinforces the predictive power of RDCC:

1. The relaxation parameter freezes early.
2. All observables depend on the same infrared value.
3. The RDCC prediction vector is stable across ten orders of magnitude in scale.
4. The global fit is dynamically justified.

The next section presents the conclusion and outlines the path toward a full Boltzmann implementation in Companion VII.

8 Conclusion

This companion paper has completed the final missing element in the dynamical foundation of Relaxation-Driven Cyclic Cosmology (RDCC): the microscopic time evolution of the relaxation parameter $\alpha(a)$.

Starting from the unique dimension-six inter-sector operator, we derived the relaxation equation

$$\dot{\alpha} = \Gamma\,\alpha, \tag{36}$$

evaluated the naive EFT estimate $\Gamma \sim H^5/M^4$, and demonstrated that it fails to generate any observable relaxation over cosmic history. This failure is structural: the suppression by powers of H/M is so extreme that the naive rate predicts $\alpha(a) \approx \alpha_{\rm min}$ for all times.

We then introduced the minimal generalized relaxation rate consistent with the CPT-symmetric structure of RDCC,

$$\Gamma = \beta H, \tag{37}$$

and showed that matching to the observed infrared value $\alpha_{\rm IR} \simeq 0.0175$ fixes the effective coupling to $\beta \simeq 1.25 \times 10^{-4}$. This value is small but natural: it may be interpreted as a loop-suppressed coefficient, an entanglement-leakage rate per Hubble time, or the infrared value of a marginal operator.

Solving the full evolution equation on a ΛCDM background revealed a simple and robust dynamical picture:

1. Near the bounce, $\alpha(a)$ grows rapidly as the Universe departs from the CPT-symmetric fixpoint.
2. By $a \sim 10^{-4}$, the relaxation parameter has reached its infrared value.
3. For all later epochs, $\alpha(a)$ is effectively frozen.

This freezing behavior explains why the RDCC prediction vector

$$(n_s,\ \Delta N_{\rm eff},\ f\sigma_8,\ \Omega_{\rm GW})$$

is stable across more than ten orders of magnitude in scale. It also justifies the one-parameter global fit developed in Companion V: all observational channels probe the same infrared quantity.

Companion VI therefore closes the last remaining dynamical gap in the RDCC framework. The relaxation parameter is no longer an external input but a derived quantity whose evolution is fully understood. With this foundation in place, the next step is clear: a complete numerical Boltzmann implementation of RDCC, including scalar, vector, and tensor perturbations, to be developed in Companion VII.

The relaxation dynamics are now under full theoretical control. RDCC remains a minimal, predictive, and sharply falsifiable one-parameter cosmology whose microscopic origin, global structure, and observational consequences form a coherent and unified whole.

Appendix A: Derivation of the Generalized Rate

The purpose of this appendix is to justify the minimal generalized relaxation rate

$$\Gamma = \beta H, \tag{38}$$

introduced in Sec. 4. We show that this form arises naturally when the dimension-six inter-sector operator is evaluated on a homogeneous FLRW background, once curvature corrections and the suppression of higher-dimensional operators are taken into account. The derivation also clarifies why the naive EFT estimate $\Gamma \sim H^5/M^4$ fails and why a linear dependence on H is the unique minimal choice consistent with the global CPT structure of RDCC.

A.1 Starting Point: The Inter-Sector Operator

The interaction Lagrangian is

$$\mathcal{L}_{\rm int} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}. \tag{39}$$

In the bipartite Hilbert-space picture of Companion IV, this operator governs the entanglement exchange between the visible and mirror sectors as the Universe departs from the CPT-symmetric bounce.

Expanding the stress-energy tensors around the background,

$$T_{\mu\nu} = \bar{T}_{\mu\nu} + \delta T_{\mu\nu},$$

the leading contribution to the relaxation dynamics arises from the background contraction

$$\bar{T}^{(+)}_{\mu\nu} \bar{T}^{(-)\mu\nu}.$$

In an FLRW spacetime,

$$\bar{T}_{\mu\nu} = \mathrm{diag}(\rho,\, p,\, p,\, p),$$

so the contraction reduces to

$$\bar{T}^{(+)}_{\mu\nu} \bar{T}^{(-)\mu\nu} = \rho_+\rho_- - 3p_+p_-.$$

A.2 Effective Relaxation Rate

The inter-sector energy-exchange term is

$$Q \propto \frac{1}{M^2}(\rho_+\rho_- - 3p_+p_-).$$

In a radiation-dominated regime,

$$\rho \propto a^{-4}, \qquad p = \frac{1}{3}\rho,$$

so the combination becomes

$$\rho_+\rho_- - 3p_+p_- = 0.$$

Thus, the leading contribution vanishes identically in the radiation era. This is a direct consequence of the conformal symmetry of radiation and the fact that the two sectors are initially maximally correlated at the CPT-symmetric bounce.

The next-to-leading contribution arises from curvature corrections to the stress-energy tensor. These corrections scale as

$$\delta T_{\mu\nu} \sim H\rho,$$

so the effective relaxation rate becomes

$$\Gamma \propto H.$$

This is the minimal curvature-sensitive generalization of the naive EFT rate and reflects the fact that the Hubble parameter is the only available curvature scale in a homogeneous FLRW background.

A.3 Suppression of Higher-Dimensional Operators

Higher-dimensional operators of the form

$$\frac{1}{M^{2+n}}\,T^{(+)}T^{(-)}H^n$$

are suppressed by powers of $H/M \ll 1$ and therefore negligible throughout the post-bounce evolution. This suppression ensures that the linear dependence $\Gamma \propto H$ is the unique minimal choice consistent with:

- the EFT expansion,
- the FLRW background,
- the suppression of higher-dimensional operators,
- and the entanglement-leakage interpretation of α.

This completes the derivation of the generalized rate.

Appendix B: Numerical Details

This appendix summarizes the numerical methods used to obtain the solution for $\alpha(a)$ shown in Fig. 1. The purpose of the numerical analysis is to confirm the analytic result derived in Secs. 4–5: the relaxation parameter grows rapidly near the bounce, freezes by $a \sim 10^{-4}$, and remains constant thereafter. This behavior is a direct consequence of the generalized rate $\Gamma = \beta H$ and the global CPT structure of RDCC.

B.1 Differential Equation

The evolution equation solved numerically is

$$\frac{d\alpha}{da} = \frac{\beta}{a}\,\alpha,$$

with $\beta = 1.25 \times 10^{-4}$, as determined in Sec. 4. This form reflects the fact that the relaxation dynamics correspond to a multiplicative decoherence process in the bipartite Hilbert space.

B.2 Cosmological Background

The Hubble parameter is given by the exact ΛCDM expression:

$$H(a) = H_0\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}.$$

We use Planck 2018 parameters:

$$H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_r h^2 = 4.2 \times 10^{-5}, \qquad \Omega_m = 0.315, \qquad \Omega_\Lambda = 0.685.$$

These values ensure that the numerical solution accurately reflects the expansion history relevant for the relaxation dynamics.

B.3 Initial Conditions

The initial conditions are chosen to lie deep inside the EFT regime:

$$a_{\min} = 10^{-30}, \qquad \alpha_{\min} = 10^{-60}.$$

These values guarantee that the early-time evolution is dominated by the generalized rate and not by the choice of initial normalization.

B.4 Numerical Method

The differential equation is integrated using:

- an adaptive Runge–Kutta solver,
- relative accuracy 10^{-12},
- logarithmic sampling in a to resolve the rapid early-time growth.

The integration range is

$$a \in [10^{-30}, 1].$$

This range covers the entire post-bounce evolution up to the present epoch.

B.5 Robustness Checks

We verified that:

- varying $a_{\min}$ by several orders of magnitude does not change the freezing behavior,
- varying $\alpha_{\min}$ by many orders of magnitude only shifts the early-time normalization,
- the freezing of $\alpha(a)$ is insensitive to the precise cosmological parameters.

These checks confirm that the freezing of $\alpha(a)$ is a structural prediction of the generalized rate $\Gamma = \beta H$, not an artifact of numerical choices.

B.6 Summary

The numerical solution confirms the analytic result:

$$\alpha(a) \to \alpha_{\mathrm{IR}} \qquad \text{for } a \gtrsim 10^{-4}.$$

This justifies the use of a single infrared parameter in the RDCC prediction vector and demonstrates that the relaxation dynamics are fully consistent with the global CPT structure of the theory.